



**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

# **LESSON PLAN**

For

## **AI Vibe Coding for Deep Learning**

Course Code: C539

Conducted by

**Tertiary Infotech Academy Pte Ltd**

UEN: 201200696W

**Version 1.1**

## DOCUMENT VERSION CONTROL RECORD

| Version Number | Effective Date of Release | Summary of Included Changes                                                   | Author         |
|----------------|---------------------------|-------------------------------------------------------------------------------|----------------|
| 1.0            | 20 August 2026            | Initial release. 4 topics, 21 hands-on labs, 2 days / 15 instructional hours. | Dr. Alfred Ang |
| 1.1            | 4 October 2026            | Course title updated to AI Vibe Coding for Deep Learning.                     | Dr. Alfred Ang |

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## Course Information

|                          |                                                                                       |
|--------------------------|---------------------------------------------------------------------------------------|
| <b>Course Title</b>      | AI Vibe Coding for Deep Learning                                                      |
| <b>Course Reference</b>  | C539                                                                                  |
| <b>Training Provider</b> | Tertiary Infotech Academy Pte Ltd (UEN 201200696W)                                    |
| <b>Duration</b>          | 2 days · 7.5 instructional hours per day (15 hours)                                   |
| <b>Daily Timing</b>      | 9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)                     |
| <b>Mode</b>              | Instructor-led, hands-on practical labs (classroom, live online via Zoom, or on-site) |
| <b>Trainer</b>           | Dr. Alfred Ang                                                                        |

## Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Explain what AI vibe coding is, set up Cursor, GitHub Copilot and Claude as PyTorch coding partners, and apply prompting patterns that produce correct deep learning code.
- LO2: Vibe code PyTorch tensor operations, computation graphs and autograd, and review and debug AI-generated PyTorch code rather than trusting it.
- LO3: Build regression and classification neural networks in PyTorch from prompts, choosing architectures, activation functions and loss functions deliberately.
- LO4: Generate training loops, optimizers and metrics from prompts, and save, load and iterate on trained models.
- LO5: Vibe code convolutional neural networks for image classification, diagnose overfitting, and apply data augmentation, regularization and transfer learning.
- LO6: Vibe code recurrent networks for sequence data, tune and evaluate them with follow-up prompts, and package a complete deep learning project.

## Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

## Course Schedule

**Day 1 — Fundamentals to first networks - set up your AI pair programmer, vibe code tensors and autograd, then build, train and save regression and classification models.**

| Time | Duration | Topic / Activity |
|------|----------|------------------|
|------|----------|------------------|

|                    |         |                                                                                                                                                                                                             |
|--------------------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:30–9:50          | 20 min  | Welcome, course introduction, ice-breaker and ground rules                                                                                                                                                  |
| <b>9:50–10:25</b>  | 35 min  | <b>Topic 1 — AI Vibe Coding for PyTorch Fundamentals (concepts + trainer demo)</b>                                                                                                                          |
| 10:25–11:15        | 50 min  | Hands-on: Lab 1: What Is AI Vibe Coding — Your First PyTorch from a Prompt; Lab 2: Setting Up Cursor, GitHub Copilot and Claude for PyTorch                                                                 |
| 11:15–11:30        | 15 min  | Tea break                                                                                                                                                                                                   |
| 11:30–13:00        | 90 min  | Hands-on: Lab 3: Prompting Patterns for Correct Deep Learning Code; Lab 4: Vibe Coding PyTorch Tensor Operations; Lab 5: Computation Graphs and Autograd with AI Assistance                                 |
| 13:00–14:00        | 60 min  | Lunch break                                                                                                                                                                                                 |
| 14:00–14:20        | 20 min  | Hands-on: Lab 6: Reviewing and Debugging AI-Generated PyTorch Code                                                                                                                                          |
| <b>14:20–14:55</b> | 35 min  | <b>Topic 2 — Vibe Coding Neural Networks (concepts + trainer demo)</b>                                                                                                                                      |
| 14:55–15:45        | 50 min  | Hands-on: Lab 7: Neural Network Architectures, Activation and Loss Functions; Lab 8: Vibe Coding a Regression Model in PyTorch                                                                              |
| 15:45–16:00        | 15 min  | Tea break                                                                                                                                                                                                   |
| 16:00–17:50        | 110 min | Hands-on: Lab 9: Vibe Coding a Classification Model with Softmax and Cross Entropy; Lab 10: Generating Training Loops, Optimizers and Metrics from Prompts; Lab 11: Saving, Loading and Iterating on Models |
| <b>17:50–18:15</b> | 25 min  | <b>Consolidation for Topic 1 and Topic 2: reading, predicting and correcting AI-generated PyTorch — shapes, losses and the training loop</b>                                                                |
| 18:15–18:30        | 15 min  | Day 1 recap and Q&A                                                                                                                                                                                         |

*Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.*

**Day 2 — Vision and sequences - vibe code a CNN image classifier, beat overfitting with augmentation and transfer learning, then forecast with an LSTM and package the project.**

| Time      | Duration | Topic / Activity                                               |
|-----------|----------|----------------------------------------------------------------|
| 9:30–9:40 | 10 min   | Day 1 review, outstanding questions and objectives for the day |

|             |         |                                                                                                                                                                           |
|-------------|---------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 9:40–10:15  | 35 min  | <b>Topic 3 — Vibe Coding Convolutional Neural Networks (concepts + trainer demo)</b>                                                                                      |
| 10:15–11:15 | 60 min  | Hands-on: Lab 12: Overview of CNNs: Convolution, Pooling and Padding; Lab 13: Vibe Coding a CNN Image Classifier                                                          |
| 11:15–11:30 | 15 min  | Tea break                                                                                                                                                                 |
| 11:30–13:00 | 90 min  | Hands-on: Lab 14: Diagnosing Overfitting with AI Assistance; Lab 15: Data Augmentation and Regularization via Prompts; Lab 16: Transfer Learning with Pre-Trained Models  |
| 13:00–14:00 | 60 min  | Lunch break                                                                                                                                                               |
| 14:00–14:35 | 35 min  | <b>Topic 4 — Vibe Coding Recurrent Networks for Sequence Data (concepts + trainer demo)</b>                                                                               |
| 14:35–15:45 | 70 min  | Hands-on: Lab 17: Overview of RNNs, LSTM and GRU; Lab 18: Vibe Coding an LSTM for Time Series Forecasting                                                                 |
| 15:45–16:00 | 15 min  | Tea break                                                                                                                                                                 |
| 16:00–17:50 | 110 min | Hands-on: Lab 19: Tuning Sequence Models with Follow-Up Prompts; Lab 20: Evaluating and Visualizing Model Performance; Lab 21: Packaging a Complete Deep Learning Project |
| 17:50–18:15 | 25 min  | <b>Project showcase: learners demo the packaged ForgeSight project and walk through the prompts that built each model</b>                                                 |
| 18:15–18:30 | 15 min  | Course recap, Q&A, next steps and close                                                                                                                                   |

*Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.*

## Lab Reference (aligned to course topics)

| Topic                                                             | Coverage            | Labs                                     |
|-------------------------------------------------------------------|---------------------|------------------------------------------|
| <b>Topic 01: AI Vibe Coding for PyTorch Fundamentals</b>          | ~29% of course time | Lab 1, Lab 2, Lab 3, Lab 4, Lab 5, Lab 6 |
| <b>Topic 02: Vibe Coding Neural Networks</b>                      | ~24% of course time | Lab 7, Lab 8, Lab 9, Lab 10, Lab 11      |
| <b>Topic 03: Vibe Coding Convolutional Neural Networks</b>        | ~24% of course time | Lab 12, Lab 13, Lab 14, Lab 15, Lab 16   |
| <b>Topic 04: Vibe Coding Recurrent Networks for Sequence Data</b> | ~23% of course time | Lab 17, Lab 18, Lab 19, Lab 20, Lab 21   |